

Ankit Tiwari

Data Science & ML Developer

 Portfolio  [github/anktw](#)  [kaggle/ankittfzd](#)  +91 789789xxxx  [at79xxxx@gmail.com](#)  Ayodhya, India

PROFILE

Data Science and Machine Learning developer with strong foundations in statistical learning, optimization, and neural networks. Builds ML systems and algorithms from first principles and develops applied AI solutions across multimodal learning and reinforcement learning.

PROJECTS

Customer Segmentation & Churn Prediction System

08/2025 – 09/2025

FastAPI, Scikit-learn, XGBoost, SHAP, MLflow

- Built an end-to-end ML system to identify high-value customers at risk of churn, enabling data-driven retention strategies.
- Deployed real-time prediction APIs serving customer segmentation and churn scoring, reducing inference latency to **<100ms**.
- Achieved **ROC-AUC of 0.87** and simulated **18%** retention improvement by targeting high-value at-risk customers.
- Integrated explainability using **SHAP** to identify key churn drivers, improving interpretability for business stakeholders.
- Simulated targeted retention strategies achieving **15–20%** potential reduction in churn for high-value segments.

RAG-Based Patient Review Question-Answering Chatbot

10/2025 – 11/2025

Generative AI + LangChain + Gradio

- An AI system that helps hospitals instantly analyze patient reviews and answer operational queries using retrieved real-world feedback.
- Embeds hospital reviews, performs vector similarity search, and uses Gemini to generate context-aware answers with a Gradio chatbot interface.
- **Achieved 92%+ relevant response accuracy**, reduced review-analysis time by **85%**.

Real-Time Sign Language Translation System

03/2026 – 04/2026

TensorFlow, Scikit-learn, FastAPI, Streamlit, Arduino (flex sensors)

- Built a real-time edge AI system for sign language translation with low-latency inference and robust signal processing.
- Built a streaming pipeline (Arduino → sliding window → LSTM inference → smoothing + confidence filtering → TTS + UI) with real-time API endpoints and gesture-to-sentence mapping
- Achieved **25ms** inference latency and **120ms** end-to-end response time with stable real-time predictions using smoothing and confidence thresholds.

MBRL World Models Planning

08/2025 – Present

PyTorch, Gymnasium / DeepMind Control Suite, VAE, RSSM,

LSTM/GRU, MCTS, CEM, TensorBoard

- Reduce sample inefficiency in reinforcement learning by enabling agents to learn environment dynamics and plan without excessive real-world interactions.
- Built an end-to-end MBRL pipeline combining VAE-based latent encoding, RSSM dynamics prediction, reward modeling, and planning (MCTS/CEM) to select optimal actions via imagined trajectories.
- Improved sample efficiency over model-free baselines (PPO/SAC) with reduced environment interactions and stable policy learning in continuous control tasks.

EDUCATION

Bachelor of Technology (Computer Engineering)

2022 – 2026

S.I.E.T Nilokheri, Haryana afw kurukshetra University, Kurukshetra 

CGPA: 7.4

SKILLS

Languages: Python, C, JavaScript, SQL

Libraries & Frameworks: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Streamlit, TensorFlow, OpenCV, FastAPI, Next.js

Tools & Systems: Git, Docker, Linux, Jupyter Notebook

COURSES

Data Science Classroom

GeeksForGeeks

08/2025 – 11/2025 | Noida, India

CERTIFICATES

- Data Science & Machine Learning (GeeksForGeeks)
- Python Programming with DS&A (YBI Foundations) [↗](#)
- Next.js App Router Fundamentals (Vercel) [↗](#)
- React Foundations (Vercel) [↗](#)

INTERESTS

Quantitative Research, Systems Programming, Open source, Ethical Hacking (Nmap, BurpSuite)